

★ WEEK 2 · PART 1 · PRACTICE

Stationarity, Worked by Hand

Check the Three Conditions, Then Find the $|\phi| < 1$ Line

Weak stationarity, the AR(1) boundary, and two cures

Part 1's gatekeeper concept, with the arithmetic shown. We check the three conditions of weak stationarity - watching a random walk's variance grow while white noise and a stable AR stay constant - then locate the $|\phi| < 1$ boundary and match each non-stationarity to its cure.

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PART 1 · APPLICATIONS

The Gate Every Model Must Pass

Case A: check weak stationarity's three conditions and watch a random walk fail as its variance grows with time.
Case B: locate the $|\phi| < 1$ boundary between a mean-reverting series and a wandering one, and match trend- and difference-stationarity to their cures. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

Stationarity is the gatekeeper

The statistical rules of the game must not change over time - constant mean, constant variance, and lag-only autocovariance.

2

$|\phi| < 1$ separates stable from wandering

An AR(1) is stationary only when its coefficient is below one in magnitude. At one, it becomes a random walk; beyond, explosive.

3

Two non-stationarities, two cures

A deterministic trend is cured by detrending; a stochastic trend, a unit root, is cured by differencing. Using the wrong one fails.

Everything here is a calculator exercise: one variance check, one boundary, one pair of cures.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a candidate process.
- Check the conditions, or the boundary.
- Then read stationarity.

What you should watch

- How the random walk's variance grows.
- How $|\phi| = 1$ marks the boundary.
- How each cure matches its trend.

The Three Conditions

Checked on three processes

Weak stationarity asks three things: a constant mean, a constant variance, and an autocovariance that depends only on the lag. We test three processes - white noise, a random walk, and a stable AR(1) - and find that the random walk fails on the second condition: its variance grows without bound as time passes

The three processes

- White noise: pure random shocks.
- Random walk: each step adds a shock.
- AR(1): a fraction of yesterday plus a shock.

What we'll do by hand

- Compute the variance of each over time.
- See which stays constant.
- Identify the one that fails.

Weak Stationarity

Three simple conditions

weak stationarity: (1) $\mathbb{E}[y_t] = \mu$, (2) $\text{Var}(y_t) = \gamma_0$, (3) $\text{Cov}(y_t, y_{t-k}) = \gamma_k$ (lag only)

The three conditions

- The mean is the same at every time.
- The variance is the same at every time.
- Autocovariance depends only on the gap.

Why it's the workhorse

- Strict stationarity is often too strong.
- These three cover what estimation needs.
- Break any one, and the tools fail.

The Variance Over Time

White noise, random walk, AR(1)

time t	1	10	50	100	verdict
white noise	1	1	1	1	constant
AR(1)	1.56	1.56	1.56	1.56	constant
random walk	1	10	50	100	grows!

White noise and the AR(1) hold a constant variance at every time - they pass. The random walk's variance is time itself, climbing from 1 to 100 - it fails the second condition. Growing variance is the classic mark of non-stationarity.

Why the Random Walk Fails

Variance equals time

random walk: $\text{Var}(y_t) = t \sigma^2 : 1, 10, 100, \dots \nearrow \infty \Rightarrow$ variance not constant

The reason

- Each step adds an independent shock.
- Variances add, so they accumulate.
- After t steps, the variance is t .

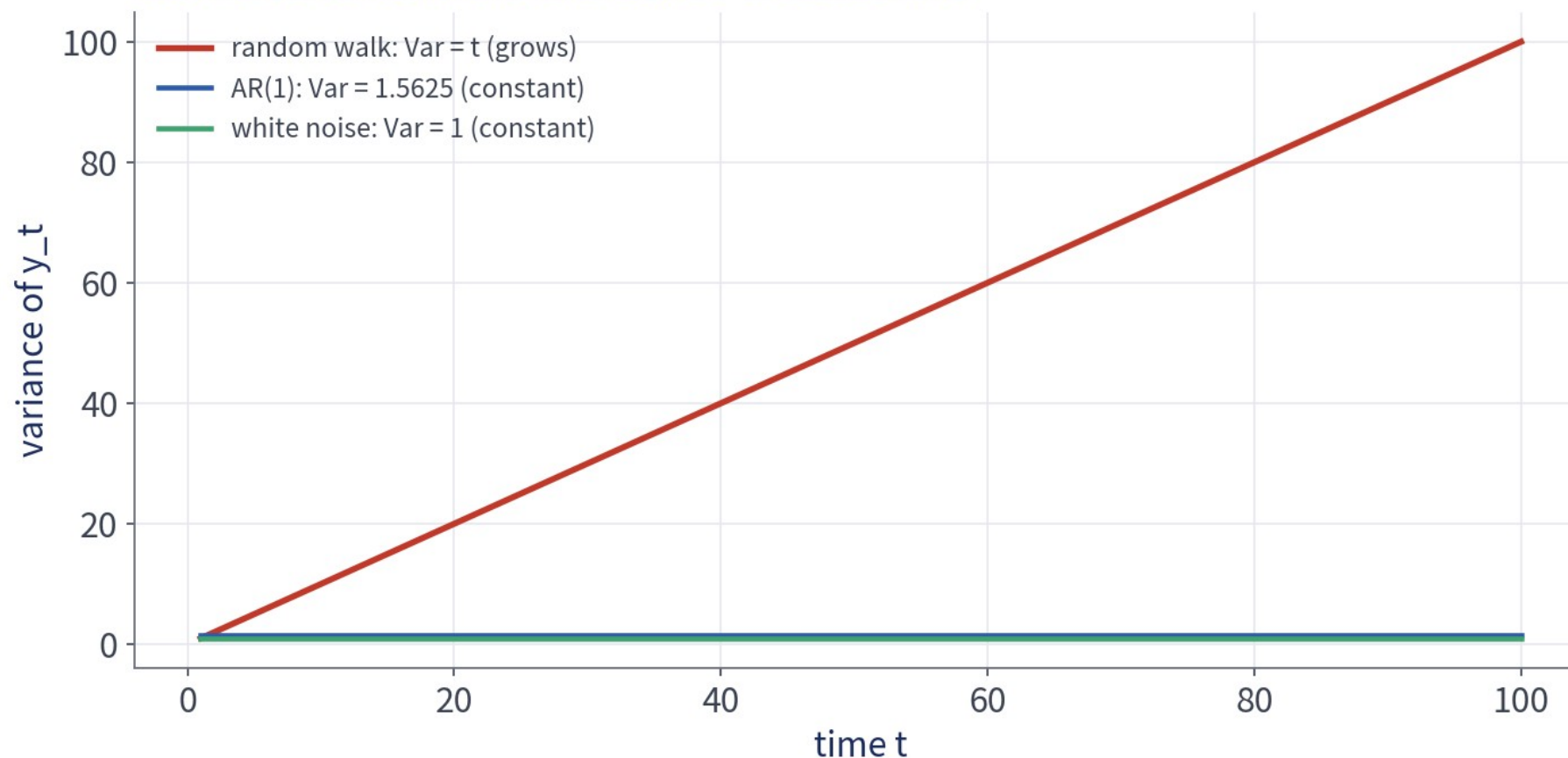
The contrast

- The AR(1) variance settles at 1.5625.
- Because the coefficient is below one.
- The past decays instead of accumulating.

One Line That Won't Stay Flat

White noise and the AR(1) hold flat variance lines - the rules of the game don't change. The random walk's variance climbs straight up, equal to time. That single diverging line is why a random walk fails stationarity, and why prices which behave this way must be transformed before modeling

Case A: only the random walk's variance grows - it fails stationarity



Why the Gate Matters

And how one sample suffices

Why stationarity first

- Estimates only make sense if the rules hold still.
- A growing variance breaks every standard tool.
- So we check it before anything else.

Ergodicity

- For a stationary series, time averages work.
- One long sample stands in for many outcomes.
- That's what lets us estimate at all.

The Boundary and the Cures

$|\phi| < 1$ and $\phi = 1$ and $\phi > 1$ are two cases
An AR(1) is stationary only when its coefficient is below one in magnitude. We compute the variance at three values - 0.6, 1.0, and 1.05 - and watch it go from finite to infinite to undefined, marking the boundary between a mean-reverting series, a random walk, and an explosive one. Then we match each kind of non-stationarity

The three regimes

- Below one: stationary, mean-reverting.
- Exactly one: a random walk.
- Above one: explosive.

What we'll do by hand

- Compute the variance at each coefficient.
- See where it breaks.
- Match trend and difference cures.

The Stationarity Condition

The line at $|\phi| = 1$

$\phi=0.6 : \frac{1}{0.64}=1.56$ (stationary) | $\phi=1 : \frac{1}{0}=\infty$ (random walk) | $\phi=1.05 : \text{explosive}$

The formula

- The variance is the shock variance over one minus phi-squared.
- It's finite only when phi is below one.
- That single condition is the whole test.

The boundary

- At phi below one, it settles.
- At exactly one, it diverges.
- Above one, it has no sensible value.

Three Regimes

Stationary, random walk, explosive

phi	1 - phi-squared	variance	behavior
0.6	0.64	1.5625	stationary, reverts
1.0	0	infinite	random walk
1.05	-0.10	undefined	explosive

At 0.6 the variance is a finite 1.5625, so the series mean-reverts. At exactly 1.0 the denominator is zero and the variance is infinite - a random walk. Above 1.0 it turns negative, meaningless: the process explodes, with shocks amplifying instead of decaying.

Two Cures for Two Trends

Detrend or difference

trend-stationary $\xrightarrow{\text{detrend}}$ stationary; difference-stationary $\xrightarrow{\text{difference}}$ stationary

Trend-stationary

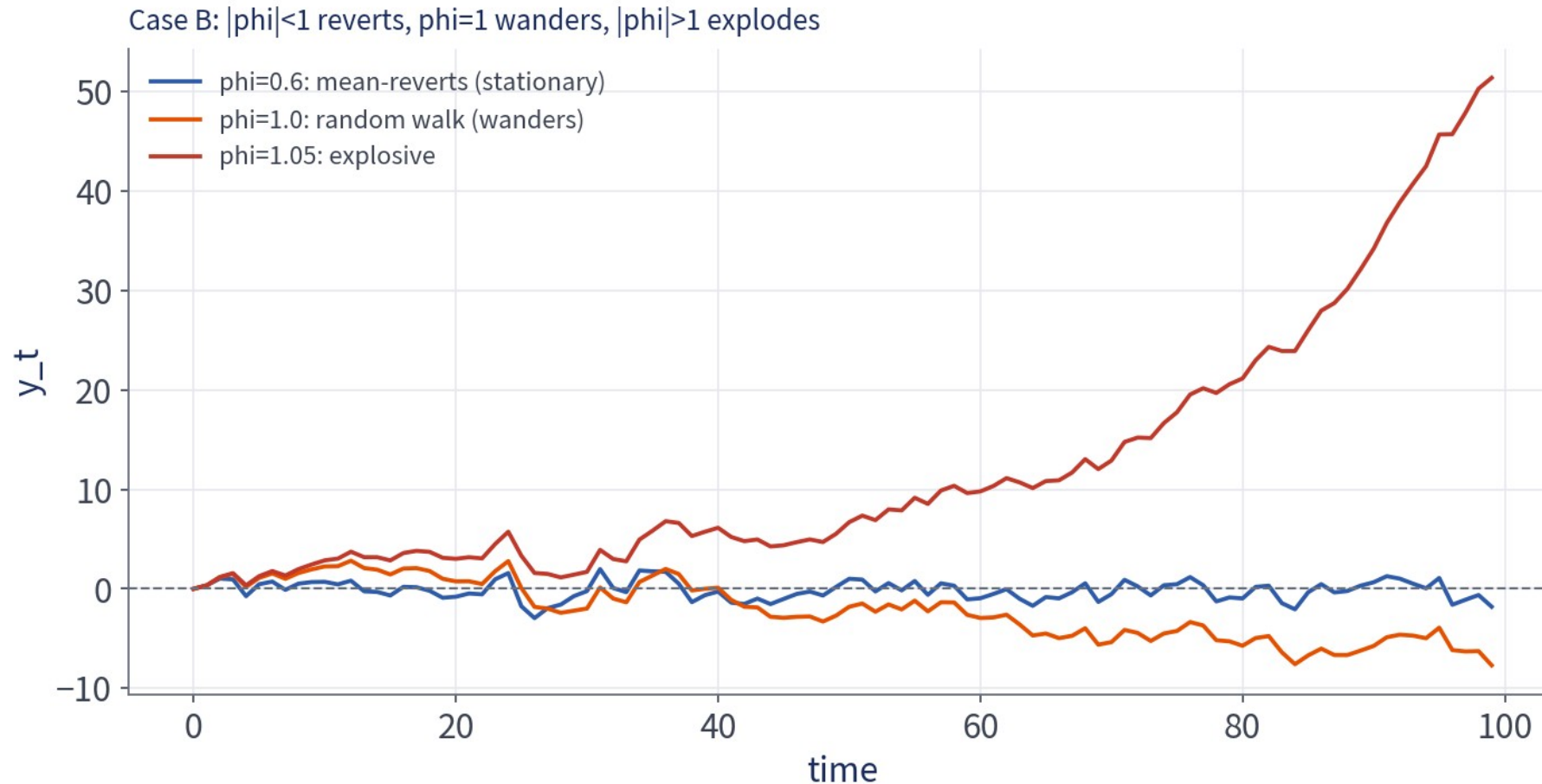
- A deterministic trend plus stationary noise.
- Cure it by detrending - regress out the trend.
- What's left is stationary.

Difference-stationary

- A stochastic trend - a unit root.
- Cure it by differencing.
- Prices are this kind; returns are the fix.

Reverts, Wanders, Explodes

Three AR(1) paths at different coefficients. At 0.6 (blue) the series is pulled back to zero again and again - stationary. At 1.0 (orange) it wanders freely with no anchor - a random walk. At 1.05 (red) it races off to infinity - explosive. The coefficient alone decides which world you're in



Mean Reversion, Made Concrete

What the boundary feels like

Below the boundary

- Shocks decay - 60% remains after one step.
- The series is pulled back to its mean.
- Half a shock is gone in about 1.4 lags.

At and above it

- At one, shocks are permanent - no reversion.
- Above one, shocks amplify and explode.
- The wrong cure leaves the trend behind.

The Stationarity Check

Before modeling at all

Stationarity is the first question you ask of any series. Check the three conditions - especially whether the variance stays constant. For an AR, confirm the coefficient is below one. And when a series isn't stationary, diagnose which kind: a deterministic trend calls for detrending, a stochastic one for differencing. Get this gate right,

or

The conditions (Case A)

- White noise and AR(1): constant variance.
- Random walk: variance equals time.
- Only the first two are stationary.

The boundary (Case B)

- ϕ 0.6 reverts, 1.0 wanders, 1.05 explodes.
- Detrend a trend, difference a unit root.
- The right cure matters.

Reproduce the Variance and the Boundary

Direct Claude Code to check your arithmetic

GOAL Show the random walk's growing variance and the AR(1) $|\phi| < 1$ boundary.

[INPUT] Prompt · Claude Code

Write and run part1_practice.py that:

CASE A - variance over time:

- * simulate white noise, a random walk, and an AR(1) with $\phi=0.6$; print the variance at $t=1, 10, 50, 100$;
- * confirm only the random walk's grows.

CASE B - the boundary:

- * for $\phi=0.6, 1.0, 1.05$, print $\sigma^2/(1-\phi^2)$ and label stationary / random walk / explosive;
- * simulate one path of each.

Then plot the variances and the three paths.

Check against the slides

- random walk $\text{Var} = t$ (1, 10, 100).
- AR(1) $\text{Var} = 1.5625$.
- $\phi=1$ diverges, 1.05 explodes.
- detrend vs difference.

Deliverable: part1_practice.py whose printed numbers match the variances and the boundary.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Which condition does a random walk break?

**Constant variance -
its variance grows
with time**

Q2 When is an AR(1) stationary?

**When the coefficient
is below one in
magnitude**

Q3 How do the two cures differ?

**Detrend a
deterministic trend,
difference a unit root**

All three come from the variance check, the $|\phi| < 1$ boundary, and the two cures.

Three Pitfalls to Avoid

Where stationarity trips people

Common mistakes

- "A constant mean is enough for stationarity."
- " ϕ just below one is basically stationary."
- "Any non-stationary series should be differenced."

The reality

- The variance must be constant too - watch it.
- Near one means near-permanent shocks - be careful.
- Detrend a deterministic trend; don't over-difference.

Two Calculations to Remember

Part 1 applications in one screen

1

The random walk fails on variance

White noise and a stable AR(1) held a constant variance - 1 and 1.5625 - at every time, passing the second condition. The random walk's variance was time itself, climbing from 1 to 100. That growing variance is the signature of non-stationarity.

2

The boundary sits at $|\phi| = 1$

At ϕ of 0.6 the AR(1) variance was a finite 1.5625, so it mean-reverts. At exactly 1.0 it was infinite - a random walk - and at 1.05 undefined, an explosive process. The single coefficient decides which regime you're in.

3

Match the cure to the trend

A deterministic trend is cured by detrending; a stochastic trend, a unit root, by differencing. Prices are difference-stationary, which is why we model returns. Getting the diagnosis right is the whole game.

Next — Part 2 practice: unit-root tests, formally distinguishing a random walk from a stationary series.